Box-Cox gamma sweep | Test All42 variants: common ICs [5, 7, 39, 44, 45, 53, 57, 64, 68, 69, 73, 76, 88]

Repaired | Cantera vs NN NRMSE | Median of IC-wise scores; completion counts use all18 ICs

Model / same-state mean	gamma	T	H2	H	O	O2	OH	H2O	HO2	H2O2	N2	10 ms status
1 ANN10	0.05	0.0366	0.397	1.6	1.02	0.219	0.992	0.166	0.988	1.14	0	18/18
1 ANN10	0.1	0.0428	0.476	1.14	1.02	0.214	1	0.157	1.03	1.23	0	18/18
1 ANN10	0.2	0.0857	0.454	0.986	1	0.559	1	0.328	0.995	1.14	0	18/18
1 ANN10	0.3	0.0553	0.379	0.863	0.753	0.24	0.925	0.0964	2.68	2.73	0	18/18
1 ANN10	0.5	0.0752	0.716	1.04	0.752	0.502	0.792	0.1	1.04	0.999	0	18/18
1 ANN10	1	0.0602	0.581	1.15	0.852	0.569	0.964	0.176	1	21.3	0	18/18
2 P-only DeepONet10	0.05	0.0487	0.18	0.931	2.21	0.28	1.01	0.141	0.964	0.999	0	18/18
2 P-only DeepONet10	0.1	0.0606	0.236	0.933	0.997	0.154	1	0.335	0.997	1	0	18/18
2 P-only DeepONet10	0.2	0.066	0.485	1	0.952	0.294	0.764	0.279	0.991	1	0	18/18
2 P-only DeepONet10	0.3	0.132	0.512	1.2	0.969	0.356	1	0.329	0.985	0.999	0	18/18
2 P-only DeepONet10	0.5	0.336	0.905	5.57	1.27	0.733	1.09	0.51	1	17	0	18/18
2 P-only DeepONet10	1	0.485	1.37	1.32	1.82	0.925	1	0.929	1.1	55.5	0	18/18
3 ANN10 + P-only DeepONet10	0.05	0.0561	0.243	0.959	2.72	0.266	1	0.132	0.97	0.999	0	18/18
3 ANN10 + P-only DeepONet10	0.1	0.0447	0.143	0.906	0.928	0.368	0.998	0.236	0.997	1	0	18/18
3 ANN10 + P-only DeepONet10	0.2	0.0442	0.448	1	0.677	0.143	0.588	0.138	0.982	0.999	0	18/18
3 ANN10 + P-only DeepONet10	0.3	0.0609	0.365	0.999	0.925	0.253	0.665	0.0997	0.96	0.996	0	18/18
3 ANN10 + P-only DeepONet10	0.5	0.107	0.573	1.61	0.995	0.594	1.01	0.175	0.897	6.07	0	18/18
3 ANN10 + P-only DeepONet10	1	0.131	0.581	1.01	0.967	0.719	0.986	0.867	1.05	15.1	0	18/18
4 T/P DeepONet10	0.05	0.291	0.413	1	1	0.637	1.07	0.356	2.27	1.83	0	18/18
4 T/P DeepONet10	0.1	0.0871	0.212	1.15	0.772	0.196	0.998	0.174	0.997	1	0	17/18
4 T/P DeepONet10	0.2	9.09e-03	0.0854	0.881	1	0.333	1	0.351	0.996	1	0	18/18
4 T/P DeepONet10	0.3	0.404	0.514	0.993	13.6	0.929	2.29	0.562	1	1	0	18/18
4 T/P DeepONet10	0.5	0.244	0.646	2.1	1.17	0.667	1.02	0.468	11.4	4.69	0	16/18
4 T/P DeepONet10	1	0.433	1.37	1	3.23	0.897	0.685	0.725	1	55.8	0	17/18
5 ANN10 + T/P DeepONet10	0.05	0.0643	0.811	1	0.952	0.471	1.04	0.244	1.93	1.96	0	18/18
5 ANN10 + T/P DeepONet10	0.1	0.0444	0.113	0.958	0.728	0.13	0.961	0.067	0.92	0.99	0	18/18
5 ANN10 + T/P DeepONet10	0.2	7.19e-03	0.122	0.744	0.71	0.252	0.548	0.148	0.903	0.998	0	18/18
5 ANN10 + T/P DeepONet10	0.3	0.0108	0.0661	0.323	0.627	0.289	0.19	0.166	0.987	0.999	0	18/18
5 ANN10 + T/P DeepONet10	0.5	0.168	0.793	1.76	1.96	0.729	1.29	0.332	9.35	5.3	0	18/18
5 ANN10 + T/P DeepONet10	1	0.275	0.893	3.32	2.47	0.729	0.599	0.256	1.34	15.4	0	18/18
6 T/P/ϕ DeepONet10	0.05	0.114	0.183	1	0.992	0.256	1.76	0.299	1.05	1	0	18/18
6 T/P/ϕ DeepONet10	0.1	0.0835	0.197	0.981	1	0.557	1	0.134	1	1	0	17/18
6 T/P/ϕ DeepONet10	0.2	0.0207	0.242	0.59	0.73	0.407	1	0.448	0.998	1	0	18/18
6 T/P/ϕ DeepONet10	0.3	0.305	0.877	1.51	1.4	0.558	1.16	0.449	0.998	1	0	18/18
6 T/P/ϕ DeepONet10	0.5	0.36	0.683	12.5	5.14	0.607	3.39	0.313	5.54	1.65	0	18/18
6 T/P/ϕ DeepONet10	1	0.29	0.847	2.2	2.69	0.819	0.632	0.355	1.09	9.41	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.05	0.213	0.707	0.999	0.998	0.397	2.87	0.511	1.03	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.1	0.0594	0.121	0.981	0.561	0.21	0.997	0.0693	0.916	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.2	0.0129	0.161	0.512	0.535	0.354	0.318	0.0562	0.944	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.3	0.0437	0.78	1	1	0.444	1	0.268	0.986	0.972	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	0.5	0.129	0.896	4.15	1.56	0.604	0.929	0.341	1.32	0.997	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	1	0.245	0.842	3.17	1.76	0.728	0.419	0.452	1.1	18.3	0	18/18

NRMSE = RMSE(Cantera,NN) / RMS(Cantera). Exact user util.py default; equal time-point weights; raw ratio without ×100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Nonzero constant N2 is valid.
Gamma is the Box-Cox exponent; soft constraint is absent. New models stop at10000 updates or150 seconds; actual endpoints are exported. Cantera + six gamma curves.